

Internship Diary

Parisara-Cycle - Green Commuter Guide

Student Name: _____

Register No.: _____

Organization/Department:

Internship Period: May 4, 2026 to May 20, 2026

Project: Parisara-Cycle Android Application

Platform: Kotlin, Jetpack Compose, Firebase-ready Android app

Project overview: Parisara-Cycle is an Android application for bicycle commuters. It helps riders sign in, choose a route, report danger zones, view nearby pit-stops, track CO2 savings, and optionally share live location with nearby riders.

Current project mode: The app supports a no-key local mode using local authentication, local danger-zone storage, mock pit-stops, demo buddy markers, local route generation, and local eco-stat storage. Firebase and Google Maps/Directions can be connected later through configuration files and API keys.

Daily Work Diary

Date	Work Summary	Outcome
May 4, 2026 Monday	Studied the Parisara-Cycle project idea and identified the main user needs: safer bicycle routes, danger-zone reporting, pit-stop support, environmental tracking, and rider profile management.	Prepared a clear feature scope for a green commuter guide app and finalized the main modules required for development.
May 5, 2026 Tuesday	Reviewed the Android project structure, Gradle setup, package organization, resources, permissions, and Compose-based application flow.	Understood the codebase structure and confirmed that the app would use Kotlin, Jetpack Compose, Material 3, repositories, ViewModels, and Android location permissions.
May 6, 2026 Wednesday	Worked on the authentication flow by studying the login and registration screen, AuthViewModel, and AuthRepository behavior for Firebase and local fallback usage.	Confirmed that users can be represented through an AppUser model and that local sign-in can support demo usage when Firebase configuration is not available.
May 7, 2026 Thursday	Analyzed and improved the application shell with bottom navigation between Map, Stats, and Profile screens using the central ParisaraCycleApp composable.	Created a simple navigation flow that allows riders to move between route planning, eco-stat tracking, and account/location controls.
May 8, 2026 Friday	Studied the map screen implementation, including the keyless local map surface, current location marker, destination selection, layer filters, and map tap/long-press actions.	Established the main map interface where riders can choose a destination, view route information, and control visible layers.
May 9, 2026 Saturday	Worked on route generation logic through DirectionsRepository. Reviewed Google Directions support and the fallback local route builder used when API keys are not configured.	Ensured the app can generate a route in demo mode and can later switch to real Google bicycle routing when keys are added.

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May 10, 2026 Sunday	Reviewed the data models used by the project, including route results, danger-zone types, pit-stop types, buddy locations, map layers, and eco-stat records.	Documented the core data structure of the app and verified that each screen has the required model support.
May 11, 2026 Monday	Implemented and reviewed danger-zone reporting logic. The app supports long-press map reporting for potholes, dangerous intersections, and blocked paths.	Completed a safety-reporting workflow where rider reports can be stored locally or sent to Firestore when Firebase is configured.
May 12, 2026 Tuesday	Reviewed pit-stop support for cycle repair and water points. Checked how pit-stop markers are displayed on the map and controlled through layer filters.	Added practical commuter support by making nearby repair and water locations visible during route planning.
May 13, 2026 Wednesday	Worked on the eco-stats feature. Reviewed trip saving, daily/monthly distance calculation, and the CO2 conversion rule of 1 km cycled = 120 g CO2 saved.	Completed environmental impact tracking so generated routes contribute to daily and monthly distance and CO2 savings.
May 14, 2026 Thursday	Reviewed the Stats screen layout and metric cards for today, this month, and current daily average. Checked formatting helpers for distance and CO2 display.	Presented eco-stat data in a readable way, helping users understand the benefit of cycling instead of using fuel-based transport.
May 15, 2026 Friday	Worked on the buddy system and live location sharing behavior. Studied nearby rider filtering, active user handling, publishing/removing locations, and the Profile screen sharing switch.	Enabled a rider-community feature where nearby riders can be shown on the map while avoiding display of the active user's own marker.
May 16, 2026 Saturday	Reviewed location permission handling through manifest permissions, permission helper utilities, and request buttons on the Map and Profile screens.	Improved app reliability by allowing route source selection and live sharing to work with permission-aware behavior and fallback messages.
May 17, 2026 Sunday	Studied Firebase integration points in the app container, including Authentication, Firestore, and Realtime Database clients, while keeping local fallback behavior available.	Prepared the project for both demo execution and future cloud-backed deployment without changing the user-facing app flow.
May 18, 2026 Monday	Reviewed UI consistency across Compose screens. Checked Material 3 cards, buttons, filter chips, icons, spacing, and screen organization for Login, Map, Stats, and Profile.	Improved the app's usability and made the interface more consistent for common rider workflows.
May 19, 2026 Tuesday	Prepared setup and user documentation covering no-key mode, optional Google Maps/Directions configuration, optional Firebase setup, permissions, and demo flow.	Created supporting documentation so the project can be built, tested, demonstrated, and extended by another developer or evaluator.
May 20, 2026 Wednesday	Completed final project review by checking the main features together: sign-in, map destination selection, route generation, danger reports, pit-stops, eco-stats, buddy markers, and profile actions.	Finalized the internship work summary and confirmed that the project demonstrates a functional green commuter guide with scope for Firebase and real routing upgrades.

Overall Work Summary

During this internship period, the Parisara-Cycle Android project was analyzed and documented as a sustainable mobility application. The work focused on understanding and organizing the app's architecture, user flow, map behavior, local fallback repositories, Firebase-ready integration points, location permissions, eco-stat tracking, and supporting documentation.

Major Outcomes

- Created a clear understanding of the complete app flow from login to route planning and profile management.
- Verified the main user-facing modules: Map, Stats, and Profile.
- Documented danger-zone reporting for potholes, dangerous intersections, and blocked paths.
- Reviewed route generation with local fallback and optional Google Directions support.
- Explained eco-stat calculation using 120 g CO2 saved per kilometre cycled.
- Covered buddy location sharing and nearby rider display behavior.
- Prepared the project for demonstration using local mode without requiring Firebase or paid API keys.

Learning Outcomes

- Gained practical understanding of Kotlin Android development using Jetpack Compose and Material 3.
- Learned how ViewModels, repositories, models, and UI screens work together in a maintainable Android app.
- Understood location permissions, route data, map markers, and local persistence using SharedPreferences.
- Learned how to design an app that can run in local demo mode while still supporting future Firebase and Google API integration.

Conclusion

The Parisara-Cycle project successfully demonstrates the concept of a green commuter guide for cyclists. It combines safety, route planning, nearby support points, environmental savings, and social riding features into a single Android application. The internship work helped convert the project into a better understood, documented, and presentation-ready application.